Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 19)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe **three** effects of noise on human health other than hearing damage.

[3 marks]

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0 1 . 2 Explain how noise can damage non-living objects.

[3 marks]

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0 1 . 3 Explain how noise can reduce the breeding success of birds.

[2 marks]

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0 1 . 4 Explain why military sonar is a threat to cetaceans.

[2 marks]

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0 2 . 1 Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

0 2 . 2 Describe what methane hydrate is and where it is found.

[2 marks]

0 2 . 3 Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

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0 3 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 3 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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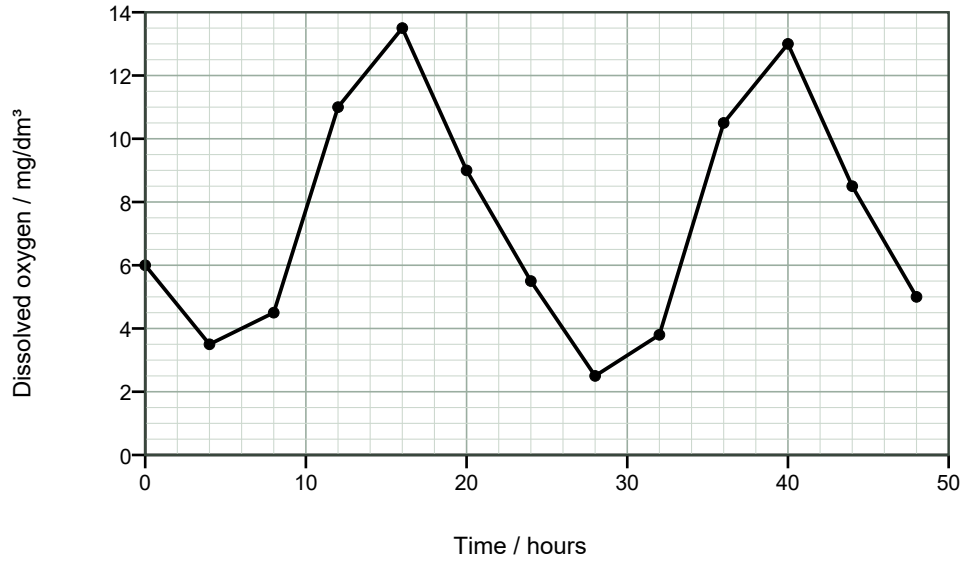
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0 4 Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



0 4 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 4 . 2 Explain the causes of the pattern shown in Figure 1.

[4 marks]

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0 4 . 3 Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

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0 4 . 4 Give **two** other harmful effects of algal blooms.

[2 marks]

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0 5 Table 1 shows the mean annual soil loss measured on four fields of one farm. Soil forms on this farm at about 1 t ha^{-1} yr^{-1}.

Table 1

Field	Slope	Land use	Mean annual soil loss / t ha^{-1} yr^{-1}
A	2%	Permanent grassland	0.8
B	9%	Maize, ploughed up and down the slope	24.0
C	9%	Maize, contour ploughed	9.0
D	1%	Spring barley, soil left bare over winter	6.5

0 5 . 1 Use Table 1 to identify the fields that are suffering **accelerated** soil erosion, and explain your answer.

[2 marks]

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0 5 . 2 Calculate the percentage reduction in soil loss achieved by changing field B to contour ploughing.

[2 marks]

Answer ☐ %

0 5 . 3 Explain how each of the following reduces erosion by water.

[3 marks]

Contour ploughing .....

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Terracing .....

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Rows of stones .....

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0 5 . 4 Field D is exposed to strong winds. Explain **four** methods, other than planting hedgerows, that would reduce wind erosion there.

[4 marks]

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0 5 . 5 Describe **four** impacts of the eroded soil **away from** the fields where it is lost.

[4 marks]

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0 6 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 6 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer _____ t/ha

0 6 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 6 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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0 7 . 1 For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases .....

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Whether the numbers of dormice differ between woodlands with different management .....

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Whether mean dissolved oxygen differs between two streams .....

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Whether crop yield increases with fertiliser application (ranked data) .....

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0 7 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 7 . 3 A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 7 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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0 8 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☒

D To measure the concentration of a pollutant at the point of release ☐

0 8 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 8 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 9 A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

0 9 . 1 Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

[5 marks]

0 9 . 2 Suggest **one** reason why such schemes may not be installed even at suitable sites.

[1 mark]

0 9 . 3 The maximum flow rate at the site is 6.5 $m^3 s^{-1}$ giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

[2 marks]

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Answer _____ kW

0 9 . 4 Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

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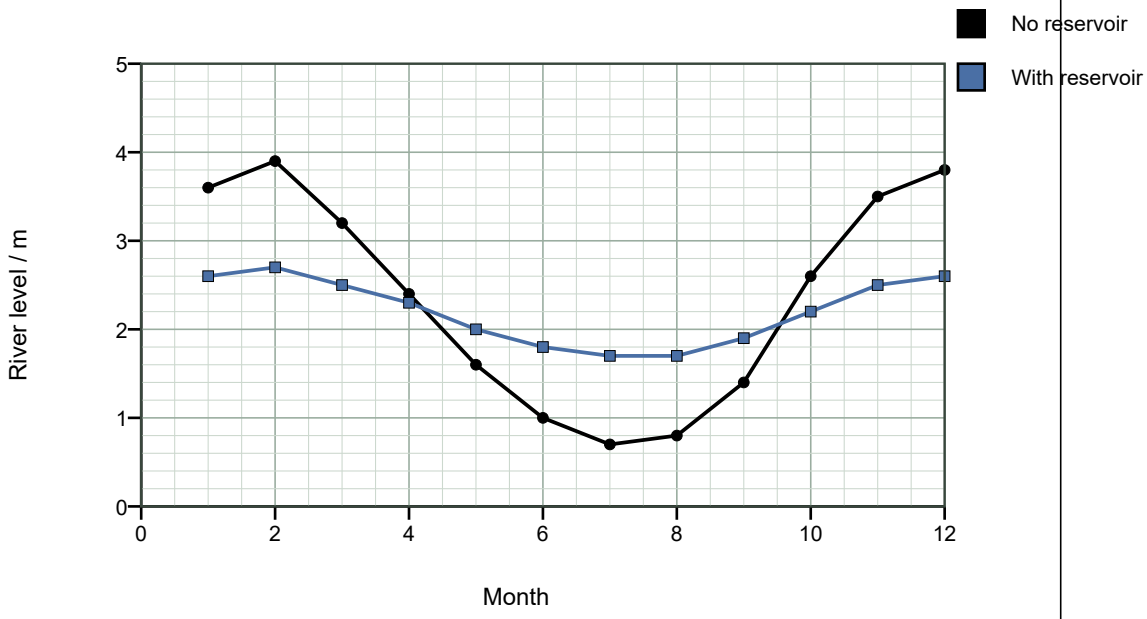
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1 0 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer [ ] m

1 0 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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1 0 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS